EV MARKET PERFORMANCE AND INSIGHTS

Tableau Story – Electric Vehicle Market Analysis

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Story Point 1: Introduction to the EV Market

Objective: To understand the overall performance of the electric vehicle market.

The electric vehicle industry is growing due to environmental awareness, government incentives, and improvements in battery technology. This Tableau Story analyzes electric vehicle sales, pricing, driving range, manufacturers, battery technologies, charging efficiency, and environmental impact.

The purpose is to transform raw market data into meaningful visual insights that support better business decisions.

Story Point 2: Overall Market Performance

Objective: To summarize the market using key performance indicators.

The Market Performance dashboard presents the major indicators of the electric vehicle market:

- Total EV sales: 1.55 billion units
- Average vehicle price: $64,921
- Average driving range: 324.54 km
- Annual CO□ savings: 738,000 tons

These indicators provide an overview of sales, affordability, vehicle capability, and environmental benefits.

Observation: The KPI dashboard gives users a quick summary of the market before they explore individual vehicle categories and manufacturers.

Story Point 3: Vehicle Category Analysis

Objective: To compare sales and performance across different vehicle categories.

The category visualization compares compact SUVs, crossovers, electric buses, electric trucks, sedans, luxury vehicles, and two-wheelers.

The analysis helps identify differences in sales contribution, driving range, and retained value across categories.

Observation: Vehicle categories have different market and performance characteristics. Luxury sedans and SUVs offer longer driving ranges, while micro city cars and two- and three-wheelers generally have shorter ranges.

Business Insight: Manufacturers can use category-level comparisons to identify suitable vehicle segments for product planning and market positioning.

Story Point 4: Manufacturer and Production Analysis

Objective: To understand the contribution of manufacturers and manufacturing countries.

The dashboard compares manufacturers using sales units and customer ratings. It also examines production output across countries.

According to the project analysis, India, China, and Vietnam lead the displayed production output.

Observation: Manufacturer comparisons help users examine differences in sales performance, customer satisfaction, and market presence.

Business Insight: Manufacturers and analysts can use these comparisons to evaluate competitive positioning and identify areas for further investigation.

Story Point 5: Driving Range and Retained Value

Objective: To compare driving range and vehicle value retention.

Driving range is an important factor in electric vehicle selection because it affects how far a vehicle can travel between charges. Retained value provides another perspective on vehicle ownership and resale considerations.

The dashboard compares range across vehicle categories and examines retained value after three years.

Observation: Luxury sedans and SUVs show longer driving ranges in the project analysis. Electric buses have the highest reported three-year retained value in the dataset.

Business Insight: Buyers and manufacturers can consider driving range and retained value together when evaluating vehicle categories.

Story Point 6: Sales Trend Over Time

Objective: To examine changes in electric vehicle sales across model years.

The sales trend visualization shows annual sales from 2018 to 2026. It helps users identify patterns over time and compare the contribution of different vehicle categories.

Observation: The project report describes annual sales as relatively stable over the displayed period, with two-wheeler scooters leading the reported sales trend.

Business Insight: Historical sales trends can support production planning, demand analysis, and future market research.

Story Point 7: Battery Technology Performance

Objective: To compare battery technologies using sales and cycle-life measures.

Battery technology influences vehicle performance, durability, and charging requirements. The dashboard examines battery technologies using the available sales and battery-life measures.

Observation: NMC and LFP batteries lead in reported sales units, while Solid-State and LFP technologies show the longest cycle life in the project analysis.

Business Insight: Comparing battery adoption and cycle life helps stakeholders understand technology preferences and evaluate potential areas for further development.

Story Point 8: Charging and Energy Efficiency

Objective: To understand the relationship between charging capability and energy consumption.

The charging and efficiency visualization compares charging power with vehicle energy consumption per kilometer.

Charging capability and efficiency are important when evaluating vehicle performance and practical charging requirements.

Observation: The project report notes that higher-capacity vehicles can charge at approximately 350 kW but may also consume more energy per kilometer.

Business Insight: Vehicle comparisons should consider both charging capability and energy efficiency rather than relying on a single performance measure.

Story Point 9: Regional Environmental Impact

Objective: To examine estimated environmental benefits across regions.

The environmental impact visualization compares regional CO□ savings using the measures available in the dataset.

Electric vehicles may help reduce emissions depending on electricity generation and the vehicles they replace.

Observation: The dashboard provides a regional comparison of estimated CO□ savings, helping users examine how environmental benefits vary across the displayed regions.

Business Insight: Regional environmental analysis can support sustainability planning and help identify areas for further investigation.

Story Point 10: Final Insights and Conclusion

The Tableau analysis brings together market performance, vehicle categories, manufacturers, driving range, retained value, sales trends, battery technologies, charging efficiency, and environmental impact.

The main findings are:

1. The KPI dashboard provides an overview of sales, average price, average range, and estimated CO□ savings.
2. Driving range and retained value differ across vehicle categories.
3. Manufacturer and country comparisons provide insight into market distribution.
4. NMC and LFP batteries lead in reported sales units.
5. Charging capability and energy consumption should be evaluated together.
6. Sales trends and regional environmental indicators provide additional perspectives on the EV market.

Conclusion: Tableau transforms electric vehicle market data into interactive visualizations that make comparisons easier and support data-driven decisions. The Story provides a guided explanation of the market and technology dashboards, helping users explore the findings in a clear and organized way.

Dashboard and Tableau Public Link

Dashboard 1: Market Performance

Dashboard 2: Technology and Impact

Tableau Public Link tableauu | Tableau Public

Project Name: Electric Vehicle Market Analysis Using Tableau